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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December 2025

IS2302-1 - USER INTERFACE DESIGN
(For IS)

Duration: 3 Hours

Max. Marks: 100

Note: Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Using sound as a signifier is helpful because:
A) It provides immediate feedback
B) It hides information
C) It removes visual cues
D) It reduces usability
2. The graphical user interface (GUI) was popularized with:
A) Command-line systems
B) Windows, icons, menus, and pointing devices
C) Assembly language
D) Punch cards
3. "Recognition rather than recall" means:
A) Interfaces should support memory by presenting visible cues
B) Users must memorize commands
C) All menus should be hidden
D) Visual clutter is encouraged
4. Error recovery should always provide:
A) Clear guidance on what went wrong and how to fix it
B) Technical codes only
C) Silent failures
D) System shutdown
5. The concept of "learnability" refers to:
A) How easily new users can begin effective interaction
B) The speed of CPU performance
C) The need for technical manuals
D) Error recovery rate
6. The term "usability" was popularized in the context of UI design to emphasize:
A) The technical efficiency of systems
B) The ease and effectiveness of use for human operators
C) The size of memory and storage
D) Programming complexity
7. Help and documentation in UI design should be:
A) Easy to search, task-focused, and concise
B) Long and complex
C) Technical only
D) Hidden from users
8. Web page design guidelines emphasize:
A) Slow navigation
B) Maintaining consistency and sense of place
C) Using only text without images
D) Minimizing accessibility
9. Long-term memory is important in UI use because:
A) It stores system code
B) It helps users recall procedures and concepts learned over time
C) It replaces icons
D) It eliminates need for training
10. Which is a disadvantage of custom controls?
A) Can improve usability for specific needs
B) May confuse users due to non-standard behavior
C) Provide familiar interaction patterns
D) Offer consistency with style guides

11. **Group boxes are used for:**
 A) Visually grouping related controls together B) Replacing menus
 C) Error reporting D) Animation
12. **Slips are errors where:**
 A) Intention is correct but execution fails B) Intention is wrong
 C) Hardware fails D) Database error
13. **Which of the following is an obstacle in the UI design process?**
 A) Clear requirements B) Conflicting user needs and business demands
 C) Standardized usability principles D) Well-defined user groups
14. **A disabled menu option should be displayed**
 A) Hidden completely B) Shown but grayed out to indicate unavailability
 C) Highlighted in red D) Displayed as an error
15. **Which heuristic deals with preventing mistakes before they occur?**
 A) Error prevention B) Consistency and standards
 C) Match between system and real world D) Help and documentation
16. **Performance measures in usability include:**
 A) Task success, error rate, completion time B) Hardware speed
 C) Disk usage D) Programming complexity
17. **Norman's "design principles" include all EXCEPT:**
 A) Feedback B) Constraints
 C) Visibility D) Redundancy elimination
18. **Poor feedback is dangerous because:**
 A) Users cannot tell if actions succeeded or failed B) It simplifies decision-making
 C) It eliminates slips D) It increases aesthetics
19. **A good design should aim to:**
 A) Make actions discoverable and results understandable B) Hide options for expert use only
 C) Increase complexity for advanced users D) Replace feedback with aesthetics
20. **A software update that prevents unsafe installation is an example of:**
 A) Lock-out forcing function B) Interlock
 C) Affordance D) Slips

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the significance of human–computer interaction in modern UI design.	8	L*2	1	1
	b) Explain with examples how usability goals and user experience goals differ.	8	L2	1	1
2.	a) Describe the phases of the requirement analysis stage in the user-interface design process.	8	L2	1	1
	b) Analyze the effect of poor requirement analysis on UI usability with an example.	8	L3	2	1
3.	a) Explain any four key human characteristics that affect interface designs.	8	L2	2	1
	b) Case Study: Evaluate the login screen of a web application using usability heuristics and suggest improvements.	8	L3	2	1
Unit – II					
4.	a) Explain menu-phrasing guidelines and how they enhance user comprehension.	8	L2	3	1

b)	Design a contextual (pop-up) menu for a word-processing application and justify its structure.	8	L3	3	1
5. a)	Differentiate between tiled and overlapping windows with examples.	8	L2	4	1
b)	Evaluate how window management influences multitasking efficiency.	8	L3	3	1
6. a)	Describe the use and importance of device-based controls in interface design.	8	L2	4	1
b)	Given a scenario of a ticket-booking kiosk, propose an effective layout of screens ensuring clear feedback and minimal user error.	8	L3	3	1
Unit – III					
7. a)	Illustrate the concept of “gulfs of execution and evaluation” with suitable examples.	8	L2	5	1
b)	Apply Norman’s seven fundamental design principles to analyze a food-delivery app.	8	L3	5	1
8. a)	Discuss how physical, semantic, and logical constraints can be combined to ensure safer product design.	8	L2	5	1
b)	Case Study: A user consistently chooses the wrong printer from a dropdown list. Identify the error type and redesign the interface to prevent it.	8	L3	5	1

BT* Bloom’s Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
